

Simran Mann

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EDUCATION

Simon Fraser University

Sept 2022 – Dec 2026

Bachelor of Applied Science in Computing Science

GPA: 3.99/4.33

- **Relevant Courses:** Distributed Systems, Compilers, Computer Architecture, Operating Systems

TECHNICAL SKILLS

Programming Languages: C/C++ (multithreading, atomics, STL, RAII), Python, x86-64 Assembly, Bash

Systems & Concurrency: POSIX threads, std::thread, mutexes, atomics, MPI, lock-free progress guarantees

Tools & Platforms: Linux/Unix, Git, Docker, LLVM, STM32

EXPERIENCE

SFU Robot Soccer

May 2025 – Present

Firmware Developer

- Worked as part of a team developing a fleet of autonomous soccer robots on ATSAME MCUs with BLDC motors, IMUs, and kicker/dribbler mechanisms, communicating with a central brain via UART and LoRa
- Designed basestation firmware to parse commands from the software over UART and relay them via SX1276 LoRa to each robot, implementing CRC error detection and extracting functionality from the datasheet to configure wireless communication
- Implemented robot movement and rotation controls using BLDC motors and coordinated dribbler, chipper, and kicker operations, ensuring real-time responsiveness.
- Gained experience debugging the sensor board over SPI while testing Mahony filter-based sensor fusion with real data
- Collaborate with the software team on continuous integration and testing to ensure robots follow instructions from an overhead camera-guided brain

NETGEAR

Sep 2024 – April 2025

Orbi Software Test Engineer

- Designed and deployed multi-node network testbeds using managed switches, configuring VLANs, static routing, and network topologies to simulate mesh network environments
- Developed Python automation tool to parse UART serial output, configure devices remotely, and automate network stability testing, reducing test execution time from 8 hours to 2 hours
- Debugged network-level issues including packet loss, routing failures, and inter-device communication problems using Wireshark, tcpdump, ping, traceroute, and iperf for bandwidth analysis
- Validated router and mesh network behavior over TCP/UDP traffic, testing packet forwarding, protocol-level correctness, latency, and failover scenarios

PROJECTS

Parallel and Distributed Graph Algorithms

Sept – Dec 2024

- Implemented parallel PageRank and Triangle Counting algorithms using std::thread for shared-memory systems and MPI for distributed multi-node clusters
- Scaled implementations from single-node multithreaded execution to distributed MPI-based execution, optimizing message-passing to reduce synchronization overhead
- Evaluated vertex-based, edge-based, and dynamic task decomposition approaches to improve load balance and inter-process communication

Concurrent Data Structures

Oct – Nov 2024

- Implemented non-blocking queues, stacks, and sets using lock-free linked lists
- Benchmarked data structures under multi-producer/multi-consumer workloads to analyze contention bottlenecks
- Addressed ABA problems via version-tagged pointers and compare-and-swap (CAS) atomic operations

C Language Compiler

June – Aug 2024

- Developed compiler for C-like language using Flex/Bison and LLVM IR, including SSA-form code generation and short-circuit evaluation for booleans
- Implemented parsing, semantic analysis, and code generation pipelines
- Built C runtime and Python scripts to automate testing, validating compilation and execution of code